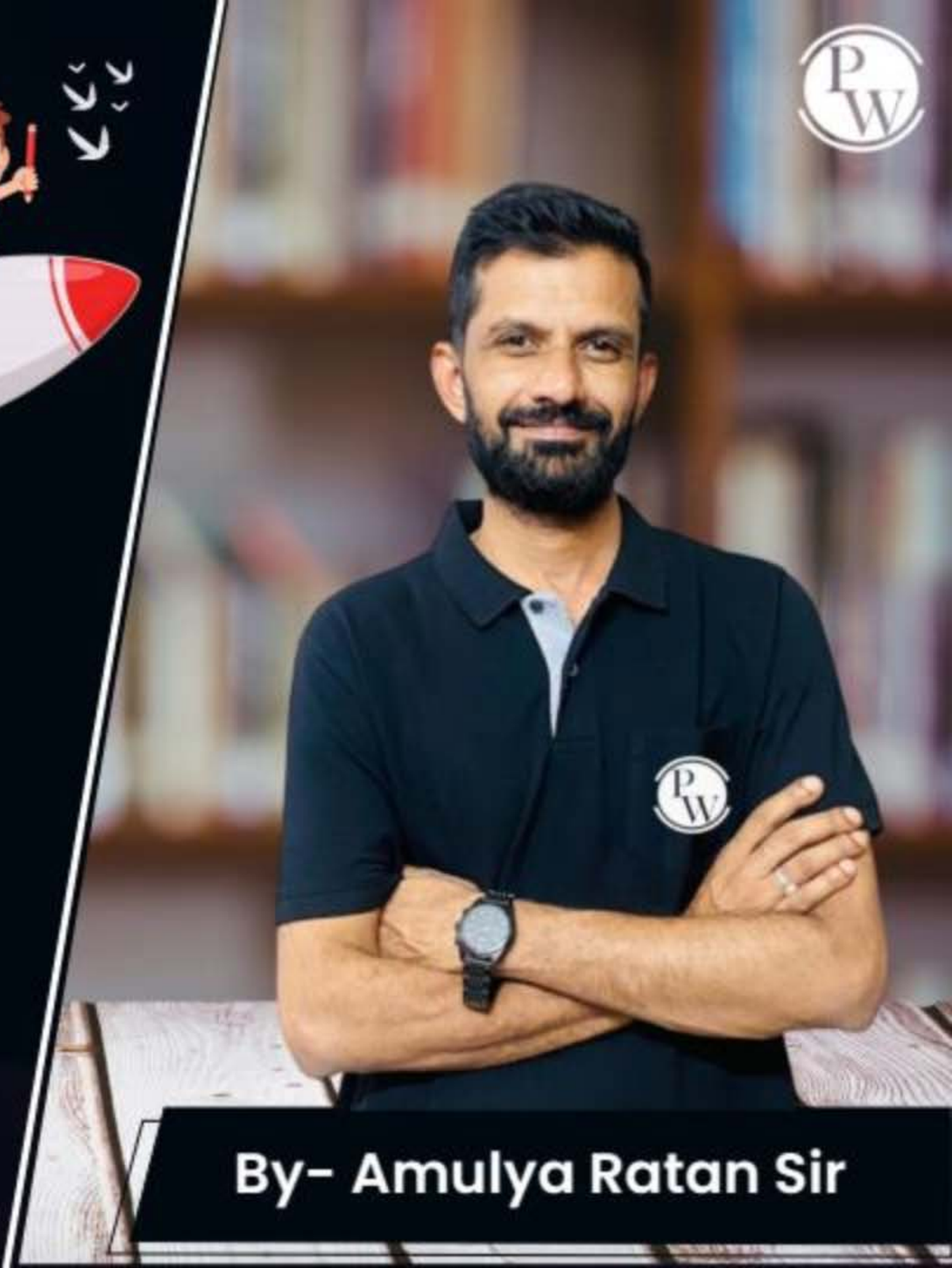
GATE

ALL BRANCHES

General Aptitude

QUANTITATIVE APTITUDE



Lecture No.- 09

By- Amulya Ratan Sir

Recap of Previous Lecture

Topic Time and Work, Pipes and Cistern

Topics to be Covered

Topic-1 Time and Distance

[MCQ]

#Q. Two pipes A and B can fill a tank in 8 minutes and 12 minutes respectively. Both the pipes are opened together and after 3 minutes, pipe A is turned off. What is the total time required to fill the tank?

Assignment

A. 3 minutes

B. 7.5 minutes

C. 10 minutes

D. 4.5 minutes

$A = \frac{1}{8}$ | $B = \frac{1}{12}$

'x' minutes

$\frac{3}{8} + \frac{x}{12} = 1$

$9 + 2x = 24$

$2x = 15$

$\therefore x = \frac{15}{2}$

$= 7.5$ min

[MCQ]

#Q. Pipes A and B can fill a tank in 4 hours and 8 hours respectively. Pipe C can empty it in 16 hours. If all the three pipes are opened together, then how long will it take to fill the tank?

Assignment

$A = \frac{1}{4}$ | $B = \frac{1}{8}$ | $C = \frac{-1}{16}$

A. 4.5 hours

B. 2 hours

C. 6 hours

D. 3.2 hours

$\frac{1}{4} + \frac{1}{8} - \frac{1}{16}$

$= \frac{5}{16} \rightarrow \frac{16}{5}$

$= 3.2$ hr

[MCQ]

#Q. A tank can be filled by 20 buckets each of capacity 13.5 litres. If the capacity of each bucket be 9 litres, how many buckets will fill the same tank?

Assignment

Tank Cap = 20 × 13.5

= 270 lit

$\frac{270}{9} = 30$

A 30

B 25

C 20

D 15

[NAT]

#Q. A tank consist of three taps A, B and C. Tap A and tap B can fill the tank in 20hrs and 12 hrs respectively, whereas tap C can empty the tank in 18hrs. C was kept open whereas A and B are opened in alternate hours i.e. 1st hr A, 2nd hr B, 3rd hr A.....so on. In how many hours the tank would be filled?

$A = \frac{1}{20}$ | $B = \frac{1}{12}$ | $C = \frac{-1}{18}$

1	2	3	4	5
A & C	B & C	A & C	B & C	A & C

55 min

30 feet

2 min → 1 feet

54 min → 27 feet

56 min → 28 ft

$\frac{1}{45}$?

1st Log

$\boxed{89\frac{4}{5} \text{ hours}}$

$A\&C = \frac{1}{20} - \frac{1}{18} = \frac{-1}{180}$

1 work B&C = 36

$\frac{1}{45}$ " = $\frac{4}{36} \times \frac{1}{45}$

~~91 hrs~~

$\frac{4}{5}$

2nd Log

$B\&C = \frac{1}{12} - \frac{1}{18} = \frac{5}{180}$

$\frac{1}{36}$

2nd & 3rd (hour) $\Rightarrow \frac{5}{180} - \frac{1}{180} = \frac{4}{180} = \frac{1}{45}$ → 2 hours

44×45 44×45

89 hrs → $\frac{44}{45}$ → 88 hrs = 1 → 90 hrs

Time & Distance

(D) Distance = Speed (S) × Time (T)

$$D = S \times T$$

$$S = \frac{D}{T}$$

$$T = \frac{D}{S}$$

Distance = 400m

Speed = 40 km/hr

Time = $\frac{400m}{40 km/hr}$ = 10 min/hrs

Time & Distance

km/hr $\xrightarrow{\times \frac{5}{18}}$ m/sec

km/hr $\xleftarrow{\times \frac{18}{5}}$ m/sec

$$\frac{km}{hr} = \frac{\cancel{1000}^{5}\,m}{\cancel{3600}_{18}\,sec}$$

Relative Speed

opposite → (+)

Same → (−)

Average Speed

[MCQ]

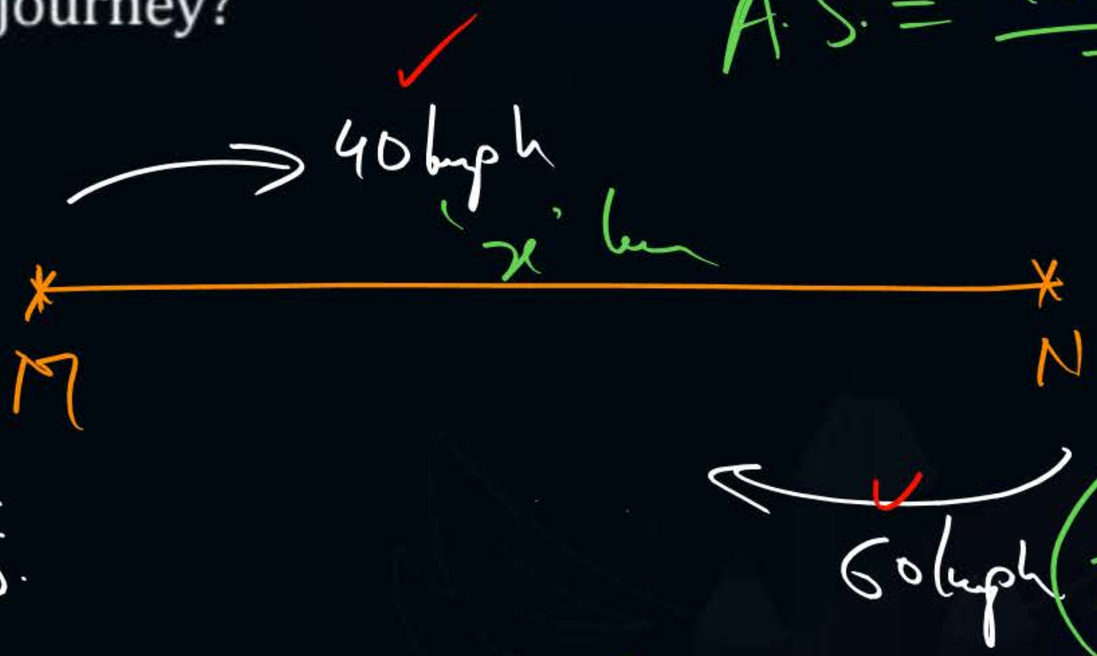
$\frac{2 \times 40 \times 60}{100} = 48$

#Q. If you travel from M to N at 40km/hr and N to M at 60 km/hr. What would be your average speed of the journey?

$\frac{2xy}{x+y}$

$A.S. = \frac{\text{Total Distance}}{\text{Total Time (T.T.)}}$

$T = \frac{D}{S}$



$T.T. = \frac{T.D}{A.S.}$

$\frac{x}{40} + \frac{x}{60} = \frac{2x}{A.S.}$

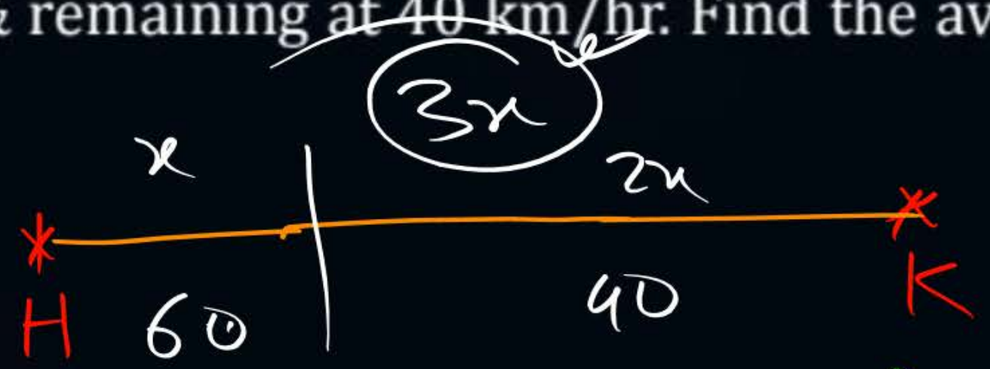
$\frac{5x}{120} = \frac{2x}{A.S}$

24

$\therefore A.S. = 48 \text{ km/hr}$

[MCQ]

#Q. A man travels from Howrah to Kanpur in his car. If he covers 1/3rd of the distance at 60 km/hr & remaining at 40 km/hr. Find the average speed of his journey.



$$\frac{x}{60} + \frac{2x}{40} = \frac{3x}{A.S.}$$

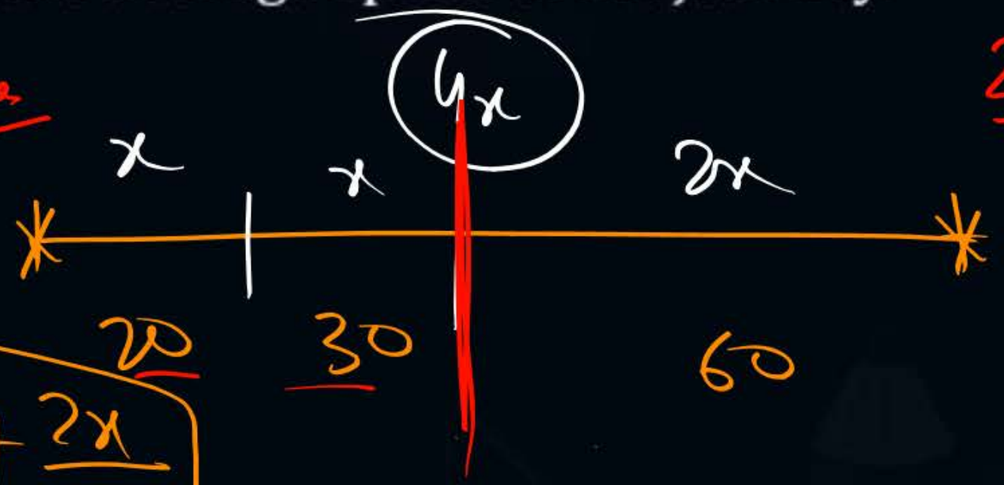
$$\frac{\cancel{8x}^{1}}{\cancel{120}_{15}} = \frac{3x}{A.S}$$

A.S = 15 × 3 = 45 km/hr

[MCQ]

#Q. A man travels ¼th of his journey at 20 km/hr, another ¼th at 30 km/hr & remaining at 60 km/hr. Find the average speed of his journey.

$\frac{2\times 20\times 30}{50} = 24\,km/hr$



$\frac{2\times 24\times 60}{84}$

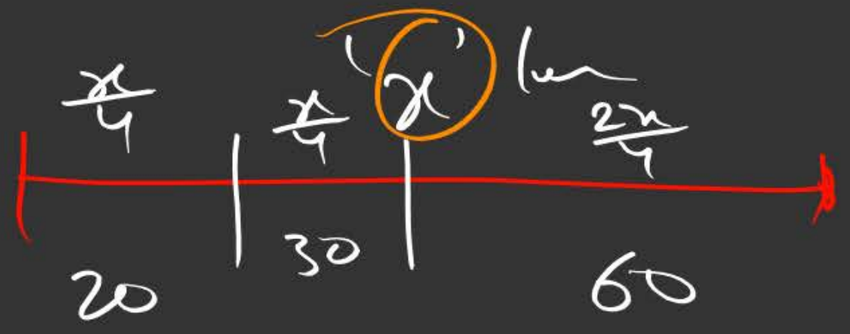
$$\boxed{\frac{4x}{A.S.}} = \boxed{\frac{x}{20}+\frac{x}{30}+\frac{2x}{60}}$$

$$\frac{7x}{60} = \frac{4x}{A.S}$$

$$A.S. = \frac{60\times 4}{7} = \frac{240}{7}$$

$= 34.28$ OR $34\frac{2}{7}$

$$T = \frac{D}{S}$$



$$\frac{x}{4 \times 20} + \frac{x}{4 \times 30} + \frac{2x}{4 \times 60} = \frac{x}{A.S.}$$

$$\frac{1}{4}\left(\frac{x}{20} + \frac{x}{30} + \frac{2x}{60}\right) = \frac{x}{A.S.}$$

$$\frac{x}{20} + \frac{x}{30} + \frac{2x}{60} = \frac{4x}{A.S.}$$

[MCQ]

#Q. If a car travels along four sides of a square at 100 kmph, 200 kmph, 300 kmph and 400 kmph. Find its average speed.

$A.S. = 48 \times 4 = 192$

$\frac{25x}{1200} = \frac{4x}{A.S}$

48

A. 152 km/hr

B. 180 km/hr

C. 192 km/hr

D. 210 km/hr

$\frac{x}{100} + \frac{x}{200} + \frac{x}{300} + \frac{x}{400} = \frac{4x}{A.S}$

$\frac{12x + 6x + 4x + 3x}{1200} = \frac{4x}{A.S}$

[MCQ]

#Q. A tourist covers half of his journey by train at 60 km/hr, half of the remainder by bus at 30 km/hr and the rest by cycle at 10 km/hr. Average speed of the tourist during the journey is

Assignment

A. 36 km/hr

B. 33 km/hr

C. 24 km/hr

D. 18 km/hr

[MCQ]

#Q. A person covers a certain distance at a certain speed. If he decrease his speed by 20%, then he takes 16 minutes more to cover the same distance. Find the time taken by him to cover the original distance at original speed.

$\boxed{U.T. = 64 \text{ minutes}}$

$\frac{\cancel{100}^{5}}{\cancel{80}_{4}} U.T - U.T = 16 \text{ minutes}$

$D = S \times T$

$\frac{80}{100}$ $\frac{100}{80}$

$\left(\frac{5}{4} - 1\right) U.T = 16$

$\frac{1}{4} U.T = 16$

[MCQ]

#Q. Walking 3/4th of the usual speed a man is late to office by 10 minutes, find the usual time.

A. 40 minutes

B. 12 minutes

C. 60 minutes

D. 30 minutes

$D = S \times T$

$\frac{3}{4}$ $\frac{4}{3}$

$\frac{4}{3} U \cdot T - U \cdot T = 10$

$\frac{1}{3} U \cdot T = 10$

$U \cdot T = 30$ minutes

[MCQ]

$\frac{8}{900} = 0.008$ $D = 0.8$ $T = \frac{0.8}{90}$

#Q. Mumbai police observes a thief 800 metres away from him. The thief started running at 80 km/hr & the police at 170 km/hr. In how much time the police can catch the thief?

$D = 800$ m

$R.S. = 90$ km/hr $\times \frac{5}{18}$

$= 25$ m/sec

$T = \frac{800 \text{ m}}{25 \text{ m/sec}}$

$= 32$ seconds

A. 8.88 minutes

B. 8.88 hours

C. 32 seconds

D. 32 minutes

[MCQ]

$Time = \frac{D}{S}$ $T = \frac{5}{6}$

#Q. An employee goes to his office from his house at a speed of 60 km/hr and reaches his office 10 minutes late. If he follows the speed of 80 km/hr, he reaches his office 5 minutes early. Find the distance he covers to reach his office.

A. 55 km

B. 45 km

C. 50 km

D. 60 km

$\frac{x}{60} - \frac{x}{80} = \frac{\cancel{15}^{1}}{\cancel{60}_{4}}$

$\frac{x}{\frac{\cancel{240}}{60}} = \frac{1}{4}$

$x = 60$

$D = S \times T$

$60\left(T + \frac{10}{60}\right) = 80\left(T - \frac{5}{60}\right)$

Distance = 60 km

60 kph → 10:10 (10:am)

80 kph → 9:55

[MCQ]

#Q. A man misses a train by 40 minutes if he travels at 30 kmph. If he travels at 40 kmph, then also he misses the train by 10minutes. What is the minimum speed required to catch the train on time?

(A) 50 km/hr

(B) 55 km/hr

(C) 60 km/hr

(D) 45 km/hr

[MCQ]

#Q. A car driver driving at a speed of 68 km/hr locates a lorry 40 meters ahead of him. After 10 seconds, the lorry is 60 meters behind him. Find the speed of the lorry.

Assignment

2 mins Summary

Topic Time & Distance

THANK - YOU